

Higher Complex Analysis. IV

Fractional linear transformations

Today?

In this lecture, we study a particular type of function: fractional linear transformations. These are easier to handle because we can use linear algebra to simplify computations.

Matrices and fractional linear transformations

Let M be a 2×2 complex matrix:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We write T_M for the associated *fractional linear transformation*:

$$T_M(z) = \frac{az + b}{cz + d}. \quad (1)$$

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First, we need to and will always assume that $(c, d) \neq (0, 0)$, otherwise the denominator is always 0.

Reducing the possibilities

If $\det M = 0$, then $(a, b) = \lambda(c, d)$, whence $az + b = \lambda(cz + d)$ and so $f(z) = \lambda$ for all $z \in \text{Domain}(f)$. So fractional linear transformations associated to matrices with determinant 0 are essentially just constant functions. Henceforth, we assume that $\det M \neq 0$.

Reducing the possibilities

$$\text{if } \det(M) = 0 \quad az + b = \lambda(cz + d) \\ T_M(z) = \frac{\lambda(cz + d)}{cz + d} = \lambda$$

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$$T_{\lambda M} = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix} = \frac{\lambda(az + b)}{\lambda(cz + d)} = T_M$$

Observe now that $T_{\lambda M} = T_M$ if $\lambda \neq 0$, and recall that $\det(\lambda M) = \lambda^2 \det(M)$. This means that when $\det M \neq 0$, if we set $M' = (\det M)^{-1/2} M$, then $\det M' = 1$ and $T_M = T_{M'}$. Therefore we may and shall restrict our attention to fractional linear transformations associated to matrices with determinant 1.

$$\text{find } M' = \lambda M \text{ st } \det(M') = 1$$

$$\det(M') = \lambda^2 \det(M) = 1 \Leftrightarrow \lambda^2 = \frac{1}{\det(M)}$$

$$\lambda = \frac{1}{\sqrt{\det(M)}} \quad \text{hence } M' = \frac{M}{\sqrt{\det(M)}}$$

$$\text{Since } T_{\lambda M} = T_M, \quad \underline{T_M = T_{M'}}$$

Domain and range

Henceforth we assume that $\det M = 1$. First we find the domain of T_M . If $c = 0$ and $d \neq 0$, then the denominator is never 0, so $T_M(z)$ is defined for all a , that is, $\text{Domain}(T_M) = \mathbb{C}$; otherwise, the denominator is nonzero when $z \neq -d/c$, whence $\text{Domain}(T_M) = \mathbb{C} \setminus \{-d/c\}$.

$$\frac{az + b}{cz + d} \quad (c, d) \neq (0, 0)$$

$$c = 0, d \neq 0 : \quad \frac{a}{d}z + \frac{b}{d} \quad z \in \mathbb{C}$$

$$\begin{cases} c \neq 0, d = 0 \\ c \neq 0, d \neq 0 \end{cases} : \quad z \neq -\frac{d}{c} \quad z \in \mathbb{C} \setminus \{-\frac{d}{c}\}$$

Inverse transformations

Now suppose that $w \in \text{Range}(T_M)$. Then

$$\begin{aligned}w = \frac{az + b}{cz + d} &\implies wcz + wd = az + b \\ &\implies z(wc - a) = b - wd,\end{aligned}$$

and as long as $wc \neq a$, this in turn implies that

$$z = \frac{b - wd}{wc - a} = \frac{(-d)w + b}{cw + (-a)} = \frac{dw - b}{(-c)w + a} = T_{M'}(w),$$

where

$$M' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad \star \quad \frac{1}{\det(M)}$$

Then M' is exactly M^{-1} , since $\det M = 1$.

Range and domain

We conclude that

$$\text{Range}(T_M) = \begin{cases} \mathbb{C} \setminus \{a/c\} & \text{if } c \neq 0 \\ \mathbb{C} & \text{if } c = 0. \end{cases}$$

$$\text{Domain}(T_M) = \begin{cases} \mathbb{C} & \text{if } c=0, d \neq 0 \\ \mathbb{C} \setminus \{-\frac{d}{c}\} & \text{if } \begin{cases} c \neq 0 & d=0 \\ c \neq 0 & d \neq 0 \end{cases} \\ \text{undefined} & \text{if } c=0, d=0 \end{cases}$$

Simplifying the range and domain

Our discussion of the domain and range had several cases; we can simplify the statements by enlarging the set of complex numbers by adding ∞ . Indeed, we define

$$f\left(-\frac{d}{c}\right) = \lim_{z \rightarrow -d/c} \frac{az + b}{cz + d} = \infty \quad \text{and} \quad f(\infty) = \lim_{z \rightarrow \infty} \frac{az + b}{cz + d} = \frac{a}{c}$$

(we will define limits formally in the next lecture), and now we can just write

$$T_M : \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{C} \cup \{\infty\}.$$

The Riemann sphere

We often call $\mathbb{C} \cup \{\infty\}$ the Riemann sphere, and write it S : we imagine a unit sphere in three dimensions with centre at 0. We can define a function $\sigma : \mathbb{C} \rightarrow S$ geometrically by joining a point p in the plane to the north pole n of the sphere by a straight line. The line will cut the sphere at n and at one other point, which we call $\sigma(p)$. Then we may think of n as being $\sigma(\infty)$. The function σ is called *stereographic projection*.

Exercise 1

Suppose that $M, N \in M_{2,2}(\mathbb{C})$. Show that $T_M T_N = T_{MN}$ (on the Riemann sphere). Deduce that the transformation T_M is bijective and that $(T_M)^{-1} = T_{M^{-1}}$.

Answer.

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Answer. Write $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then we compute

$$MN = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

Answer to Exercise 1

Further,

$$\begin{aligned} T_M T_N(z) &= \frac{aT_N(z) + b}{cT_N(z) + d} = \frac{a\frac{ez + f}{gz + h} + b}{c\frac{ez + f}{gz + h} + d} \\ &= \frac{a(ez + f) + b(gz + h)}{c(ez + f) + d(gz + h)} = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)}. \end{aligned}$$

Answer to Exercise 1

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This formula holds for all $z \in \mathbb{C}$ for which none of the denominators is 0, and then extends to all $z \in S$ by continuity. It follows that

$$T_M T_N(z) = T_{MN}(z) \quad \forall z \in S.$$

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$$\begin{aligned}T_M T_N(z) &= \frac{aT_N(z) + b}{cT_N(z) + d} = \frac{a\frac{ez + f}{gz + h} + b}{c\frac{ez + f}{gz + h} + d} \\&= \frac{a(ez + f) + b(gz + h)}{c(ez + f) + d(gz + h)} = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)}.\end{aligned}$$

This formula holds for all $z \in \mathbb{C}$ for which none of the denominators is 0, and then extends to all $z \in S$ by continuity. It follows that

$$T_M T_N(z) = T_{MN}(z) \quad \forall z \in S.$$

In particular, $T_M T_{M^{-1}} = T_I$, and T_I is the identity mapping. Hence T_M is invertible, with inverse $T_{M^{-1}}$. △

Factorisations of fractional linear transformations

Matrices may be factorised, and hence fractional linear transformations may be factorised too.

Theorem

Every 2×2 complex matrix with determinant 1 may be written as a product of at most three matrices of the following special types:

$$\frac{a}{d}z + \frac{b}{d} \quad \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot -\frac{1}{z} \quad \text{inversion}$$

affine map

Proof. Consider the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We consider three cases, according to whether any of a or c is 0. Since the determinant is 1, both cannot be 0. $ad - bc \neq 0$ & $(c, d) \neq 0$
If $c = 0$, the matrix itself is of the desired form.

Case 1. If $c = 0$, b doesn't matter, $d \neq 0$ then $a \neq 0$

Proof of theorem

Now suppose that $a = 0$. Then

$$\begin{aligned} \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} &= \overbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1}}^{I_{2 \times 2}} \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -c & -d \\ 0 & b \end{pmatrix}, \end{aligned}$$

which is of the desired form.

Proof of theorem (continued)

Finally, if neither a nor c is 0, then we take $x = a/c$, and write

$$\begin{aligned}\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a - cx & b - dx \\ c & d \end{pmatrix};\end{aligned}$$

since $a - cx = 0$, combining with the previous case shows that the factorisation holds in this case too. $\square$

$$\begin{aligned}&\begin{pmatrix} c & a \\ 0 & c \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & b - \frac{ad}{c} \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} c & a \\ 0 & c \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -c & -d \\ 0 & b - \frac{ad}{c} \end{pmatrix}\end{aligned}$$

Proof of theorem (continued)

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This factorisation simplifies a number of arguments.

Fractional linear transformations, lines and circles

Theorem

Let T_M be a fractional linear transformation. Then the image of a line under T_M is a line or a circle, and the image of a circle under T_M is also a line or a circle.

Proof. If M is of the form $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$, then T_M is an affine transformation, and the theorem holds in this case. Otherwise, T_M is composed of some affine transformations and the *inversion* map $z \mapsto -1/z$; it therefore suffices to treat the inversion map.

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We may write the equation of the circle with centre c and radius r in the form $|z - c|^2 = r^2$. This is the same as

$$|z|^2 - 2 \operatorname{Re}(z\bar{c}) + |c|^2 - |r|^2 = 0.$$

Proof of theorem

$$|z|^2 - 2 \operatorname{Re}(z\bar{c}) + |c|^2 - |r|^2 = 0.$$

If we set $w = -1/z$, then we find that

$$\left| \frac{1}{w} + c \right|^2 \Leftrightarrow \left| \frac{-1}{w} - c \right|^2 = r^2,$$

whence

$$|wc + 1|^2 = r^2 |w|^2,$$

and

$$(|c|^2 - r^2) |w|^2 + 2 \operatorname{Re}(wc) + 1 = 0,$$

which is the equation of a circle, unless $|c|^2 = r^2$, when we get the equation of a straight line.

$$|w|^2 + \frac{2 \operatorname{Re}(wc)}{|c|^2 - r^2} + \frac{1}{|c|^2 - r^2}$$

$$\Leftrightarrow \left| w - \left(-\frac{c}{|c|^2 - r^2} \right) \right|^2 = \frac{r^2}{(|c|^2 - r^2)^2}$$

$$|wc|^2 - 2 \operatorname{Re}(wc) + 1 = r^2 |w|^2$$

$$2 \operatorname{Re}(wc) = -1$$

$$\operatorname{Re}(x+iy)(a+bi) = ax - by = -\frac{1}{2}$$

$$ax - by + \frac{1}{2} = 0$$

Proof of theorem. 2

The argument to show that the inversion mapping sends straight lines to straight lines or circles is similar, and we omit it; the starting point is that the equation of a straight line may also be written

$$|z - p| = |z - q|. \quad \square$$

$$|z - p|^2 = |z - q|^2$$

$$(|z|^2 - 2\operatorname{Re}(z\bar{p}) + |p|^2 = |z|^2 - 2\operatorname{Re}(z\bar{q}) + |q|^2$$

$$2\operatorname{Re}(z\bar{p}) - 2\operatorname{Re}(z\bar{q}) = |q|^2 - |p|^2$$

Exercise 2

Show that the fractional linear transformation $T : z \mapsto \frac{z - i}{z + i}$ sends the upper half plane onto the unit disc $\{w \in \mathbb{C} : |w| < 1\}$. Find the image under T of the wedge $\{z \in \mathbb{C} : \varphi < \text{Arg}(z) < \pi - \varphi\}$.

Answer.

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Answer. We know that T is invertible and may check easily that T^{-1} is the fractional linear transformation

$$w \mapsto -i \frac{w+1}{w-1}.$$
$$|w| < 1$$

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Answer. We know that T is invertible and may check easily that T^{-1} is the fractional linear transformation $w \mapsto -i \frac{w+1}{w-1}$.

Suppose that $z = x + iy$, where $y > 0$. Then

$$|Tz|^2 = \left| \frac{x+iy-i}{x+iy+i} \right|^2 = \frac{x^2 + (y-1)^2}{x^2 + (y+1)^2} < 1,$$

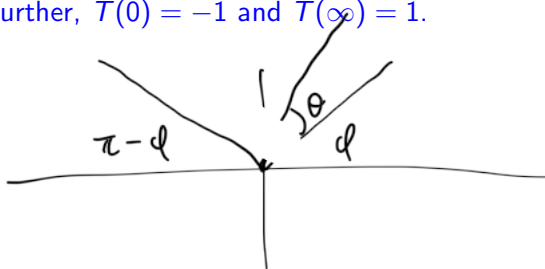
so T maps into the unit disc.

Answer to Exercise 2

On the other hand, if $|w| < 1$, then

$$\begin{aligned} T^{-1}w &= -i \frac{w+1}{w-1} = -i \frac{(w+1)(\bar{w}-1)}{(w-1)(\bar{w}-1)} \\ &= \frac{i(w - \bar{w}) + i(1 - w\bar{w})}{|w-1|^2}, \end{aligned}$$

so $\text{Im}(T^{-1}w) = (1 - |w|^2)/(|1 - w|^2) > 0$; this implies that w is in the image of the upper half plane under T , so T maps onto the unit disc. Further, $T(0) = -1$ and $T(\infty) = 1$.



Answer to Exercise 2 (continued)

Let R_θ denote the ray $\{re^{i\theta} : r \in \mathbb{R}^+\}$. Each ray R_θ is mapped into the part of a line or a circle that lies inside the unit disc and starts at -1 and end at 1 .

$$z \mapsto \frac{z - i}{z + i}$$
$$= \frac{re^{i\theta} - i}{re^{i\theta} + i}$$

Answer to Exercise 2 (continued)

Let R_θ denote the ray $\{re^{i\theta} : r \in \mathbb{R}^+\}$. Each ray R_θ is mapped into the part of a line or a circle that lies inside the unit disc and starts at -1 and end at 1 . Further,

$$\begin{aligned}Te^{i\theta} &= \frac{e^{i\theta} - i}{e^{i\theta} + i} = \frac{e^{i\theta} - e^{i\pi/2}}{e^{i\theta} + e^{i\pi/2}} \\&= \frac{e^{i\theta/2 - i\pi/4} - e^{i\pi/4 - i\theta/2}}{e^{i\theta/2 - i\pi/4} + e^{i\pi/4 - i\theta/2}} = i \tan(\theta/2 - \pi/4).\end{aligned}$$

It follows that the image of the ray is A_θ , the arc of the circle with centre $(0, \tan(\theta/2))$ between $(-1, 0)$ and $(1, 0)$.

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$$\begin{aligned}Te^{i\theta} &= \frac{e^{i\theta} - i}{e^{i\theta} + i} = \frac{e^{i\theta} - e^{i\pi/2}}{e^{i\theta} + e^{i\pi/2}} \\&= \frac{e^{i\theta/2 - i\pi/4} - e^{i\pi/4 - i\theta/2}}{e^{i\theta/2 - i\pi/4} + e^{i\pi/4 - i\theta/2}} = i \tan(\theta/2 - \pi/4).\end{aligned}$$

It follows that the image of the ray is A_θ , the arc of the circle with centre $(0, \tan(\theta/2))$ between $(-1, 0)$ and $(1, 0)$.

Hence the image of the wedge is the “eye-shaped” region between the arcs A_φ and $A_{\pi-\varphi}$. △

Exercise 3

Suppose that $M \in M_{2,2}(\mathbb{R})$ (all the coefficients of the matrices are real) and $\det(M) = 1$. Show that if $\operatorname{Im}(z) > 0$, then $\operatorname{Im}(T_M z) > 0$. Deduce that T_M maps the upper half plane $\{z \in \mathbb{C} : \operatorname{Im}(z) > 0\}$ onto itself bijectively.

Answer.

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Answer. Suppose that $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbb{R}$.

Then

$$\begin{aligned} T_M(z) &= \frac{az + b}{cz + d} = \frac{(az + b)(cz + d)^-}{(cz + d)(cz + d)^-} \\ &= \frac{(az + b)(c\bar{z} + d)}{|cz + d|^2} = \frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2}. \end{aligned}$$

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Answer. Suppose that $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbb{R}$.

Then

$$\begin{aligned} T_M(z) &= \frac{az + b}{cz + d} = \frac{(az + b)(cz + d)^{-}}{(cz + d)(cz + d)^{-}} \\ &= \frac{(az + b)(c\bar{z} + d)}{|cz + d|^2} = \frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2}. \end{aligned}$$

Suppose that $z = x + iy$, where $y > 0$. Then ...

Answer to Exercise 3

$$\operatorname{Im}(T_M(z)) = \operatorname{Im}\left(\frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2}\right)$$

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Thus $\operatorname{Im}(T_M(z)) > 0$, as required.

Answer to Exercise 3 (continued)

From a previous exercise, T_M is bijective. It remains to show that if $\operatorname{Im}(w) > 0$, then the complex number z such that $T_M(z) = w$ satisfies $\operatorname{Im}(z) > 0$. But $T_M(z) = w$ is the same as $z = (T_M)^{-1}w = T_{M^{-1}}w$; by linear algebra, $M^{-1} \in M_{2,2}(\mathbb{R})$ and $\det(M^{-1}) = 1$. The first part of this answer, applied to M^{-1} rather than M , shows that $\operatorname{Im}(T_{M^{-1}}w) > 0$, and thus $\operatorname{Im}(z) > 0$, as required. $\triangle$

Further exercises

Exercise

Suppose that M is a 2×2 complex matrix of determinant 1, and that T_M maps the upper half plane onto itself. Show that M is a real matrix.

Exercise

Find a characterisation of all 2×2 complex matrices M of determinant 1 such that T_M maps the unit disc onto itself.

Exercise

Suppose that T_M is a fractional linear transformation that maps the circle C_1 onto the circle C_2 . When does T_M map the interior of C_1 onto the interior of C_2 ?